ABC Engineering Services

Phase 1B Improvements

Overview

A number of technical issues were identified during the initial implementation of the ABC Engineering Services invoice management API. These issues did not prevent the implementation of the key functionality, so it was decided that they would not be addressed immediately. Instead, they were recorded in the project backlog for subsequent consideration.

Emerging from this was the idea of having a limited Phase 1B enhancements cycle. Issues that could be addressed within a limited period were selected for further work.

The objective was to improve code consistency, the development workflow, application diagnostics, and REST API error handling, while investigating further improvements to persistence and response handling.

The work also provided an opportunity to evaluate the scope and complexity of potential changes before committing to their implementation, keeping the enhancement cycle manageable.

Completed Enhancements

1. Supplier Reactivation Return Type

Estimate: 15–30 minutes
Actual: 45 minutes
Status: Completed

Reviewed the service method's return type and the associated controller and tests. Also aligned the unit tests with the conventions used elsewhere in the project.

2. Service Method Naming Consistency

Estimate: 15–30 minutes
Actual: 30 minutes
Status: Completed

Standardised service method and request DTO naming where appropriate, updating dependent controller code and tests. Ran the relevant tests to verify that the changes preserved existing behaviour.

3. JUnit Run Configuration Template

Estimate: 15–30 minutes
Actual: 30 minutes
Status: Completed

Created a reusable IntelliJ IDEA JUnit run configuration to simplify execution of the test suite with the required database password environment variable supplied.

4. Application Logging

Estimate: 2-4 hours
Actual: 5 hours, including one hour of tutorial time
Status: Completed

Introduced application logging using SLF4J with Logback, providing a more appropriate basis for diagnosing application behaviour and investigating problems than relying on console output alone.

5. API Error Response Standardisation

Estimate: 2-4 hours
Actual: 5 hours, including one hour of tutorial time
Status: Completed

Standardised REST API error responses using Spring's ProblemDetail support, improving the consistency and structure of error information returned by the API.

Investigated and Deferred Enhancements

6. Open EntityManager in View (OSIV)

Estimate: 1-2 hours for investigation; implementation effort not specified
Actual: Included in approximately 6 hours spent investigating OSIV and DTO/API response handling
Status: Investigated; configuration change deferred

Investigated the implications of Open EntityManager in View and its relationship with entity loading and REST response serialisation.

Disabling OSIV in the existing implementation resulted in LazyInitializationException errors, demonstrating that the current response handling depended on the persistence context remaining available during serialisation.

A configuration change alone was therefore insufficient. The issue was considered alongside the broader review of entity and DTO response handling.

7. DTO Mapping and REST Response Review

Estimate: 1-2 hours for investigation; implementation effort not specified
Actual: Included in approximately 6 hours spent investigating OSIV and DTO/API response handling
Status: Investigated; implementation deferred

Reviewed the handling of entity objects in API responses, including the risk of exposing persistence-related properties and triggering unwanted relationship serialisation or lazy loading.

Applied @JsonIgnore as an interim measure to prevent unwanted serialisation in the affected area. This addressed the immediate issue without requiring a wider redesign.

A more comprehensive solution would involve explicit response DTOs, appropriate mapping within service transactions, and a review of OSIV configuration. This would also require corresponding controller and test changes.

The full solution was deferred because its implementation and testing effort extended beyond the intended scope of this enhancement cycle. The interim measure was retained, with the broader response-handling changes identified as future work.